

ECON 164: Theory of Economic Growth

Week 6: Ideas and Innovation

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Spring 2026

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 - pure externality model: $B = AK^{1-\alpha} \rightarrow$ collapses to vanilla AK
- but, at best, innovation is **accidental** $\rightarrow$ what if it's **deliberate**?

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 - share aggregate dynamics in which A grows endogenously...
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 - ... b/c firms invent **better** varieties of goods ([Aghion and Howitt, 1992](#)) [Week 7]

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- **disclaimer:** these are models of “frontier” economies
 - discuss how technology diffuses to the rest of the world [Week 8]

The Idea Diagram

Ideas → Nonrivalry → Increasing returns → Imperfect competition

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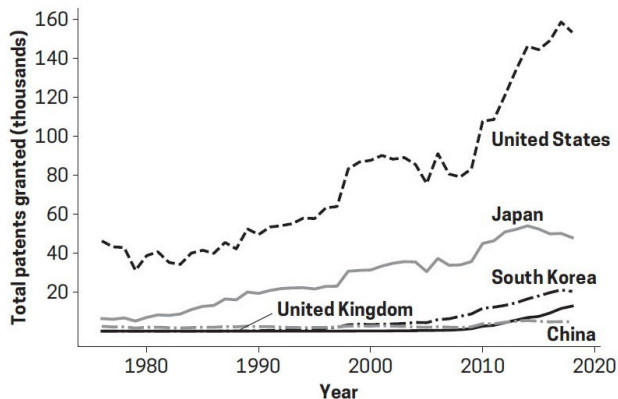
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there are endless combinations! ([Weitzman, 1998](#))

One way to see *some* idea accumulation

Figure 4.1 Patents Issued in the United States, by Country of Origin



SOURCE: OECD Patent Statistics (2014).

NOTE: These are counts of patents granted by the U.S. Patent Office, by the applicants' country of residence.

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Ideas → Nonrivalry → ...

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- Factors and goods are rival...
 - If I ate an apple, you cannot eat the same apple!
 - If you work an hour as a cook, you are not also driving an Uber that hour
 - If you use equipment to produce cars, cannot use the same to produce airplanes

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 - If you work an hour as a cook, you are not also driving an Uber that hour
 - If you use equipment to produce cars, cannot use the same to produce airplanes
- ... but ideas are **nonrival**
 - If I learn how to fish, I can show you how to fish and we can both do it
 - If I learn *once* how to make soccer balls more efficiently, I can use it forever

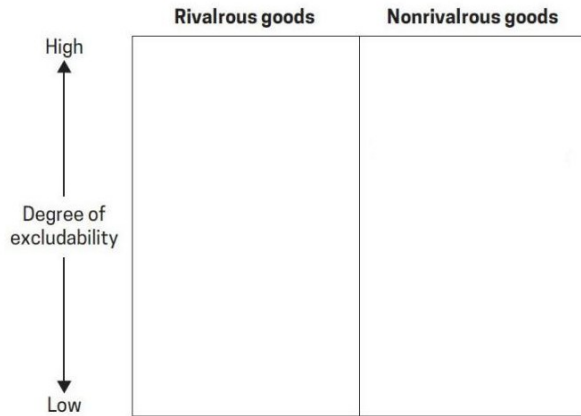
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Figure 4.4 Economic Attributes of Selected Products



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High	Legal services Smartphone	Encrypted TV show
		Open source software
		Walmart operations manual
Low	Fish in the ocean Bee pollination	National defense basic R&D calculus

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Patents make nonrival ideas excludable!

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Suppose our plant uses a standard neoclassical production function. . .

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... which exhibits constant returns to scale in $\{K, L\}$

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...the **nonrival** ideas (A) can be shared across plants!

$$F(A, \lambda K, \lambda L) = \lambda Y \quad \rightarrow \quad F(\lambda A, \lambda K, \lambda L) > \lambda Y$$

... → Increasing returns → Imperfect competition

If firms take prices as given in perfectly competitive markets...

$$\begin{aligned}\pi &= F(A, K, L) - rK - wL - fA \\ &= F(A, K, L) - \frac{\partial F}{\partial K}K - \frac{\partial F}{\partial L}L - \frac{\partial F}{\partial A}A \\ &= \underbrace{0}_{\text{b/c CRS in } \{K, L\}} - \frac{\partial F}{\partial A}A \\ &< 0\end{aligned}$$

...then they'd earn **negative profits** if they had to pay for A

Fixed costs capture increasing returns + (need for) imperfect competition

Suppose you run a pharmaceutical company designing a new drug. . .

- you pay a **one-time cost** F to do the R&D
- you pay a **marginal cost** c to manufacture and package the pills

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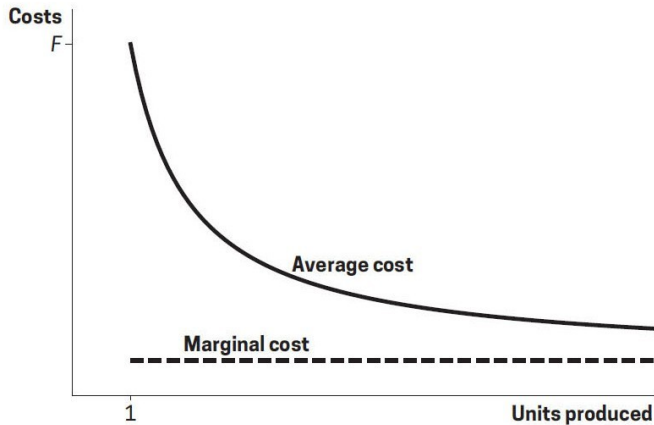
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. . . but if perfect competition, $p = c$ and thus $V = 0 \rightarrow$ need some **market power**

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Figure 4.5 Costs Functions with Increasing Returns



NOTE: Average costs decline as production increases, but never fall below marginal cost, meaning the firm can never earn positive profits.

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- embed the Idea Diagram in a general equilibrium model
- start with aggregate dynamics like we did with NGM, then microfoundations
 - aggregate dynamics are shared by many of these models, inc. next week's
 - microfoundation: firms invent new varieties of goods

Production, capital, and labor

The production function is

(with Harrod-neutral productivity)

$$Y_t = K_t^\alpha (A_t s_Y L_t)^{1-\alpha}$$

where s_Y is the share of labor allocated to **production**

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Capital and labor accumulation as in the **Solow** model:

$$\dot{K}_t = s_I Y_t - \delta K_t$$

$$\dot{L}_t = g_L L_t$$

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- $\phi > 0$: having ideas makes producing more ideas **easier**
(“standing on the shoulders of giants” w/ calculus, CRISPR, ...)
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(“fishing out” best ideas, takes longer to reach research frontier)

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The original **Romer (1990)** model has $\lambda = \phi = 1$, but we’ll study the general case

later: what does the data imply re: λ and ϕ ?

Growth of productivity

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$$g_A \equiv \frac{\dot{A}_t}{A_t} = \theta s^{\text{orange}} R^{\text{pink}} \left(\frac{L_t^{\text{pink}}}{A_t^{1-\phi^{\text{teal}}}} \right)$$

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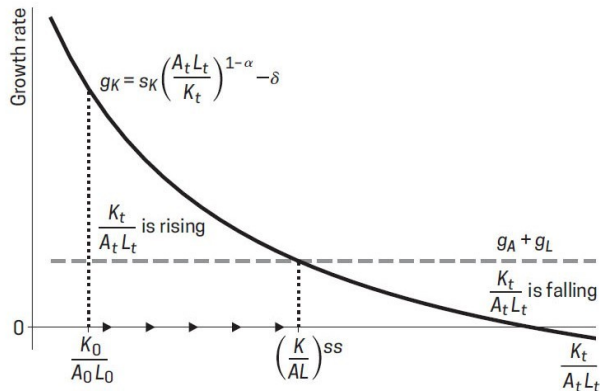
how do we find it?

Recall from the Solow model a **similar type of expression**, but for g_K :

$$g_K \equiv \frac{\dot{K}_t}{K_t} = s_I \left(\frac{A_t L_t}{K_t} \right)^{1-\alpha} - \delta$$

Use a phase diagram

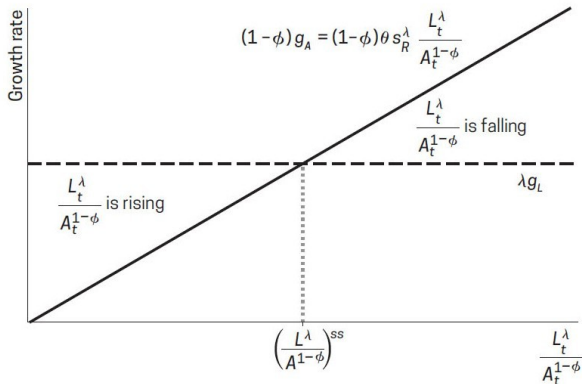
Figure 2.2 The Dynamics of the $K_t/A_t L_t$ Ratio



NOTE: The dark line plots the growth rate of capital, g_K , from equation (2.11) against the $K_t/A_t L_t$ ratio. The growth rate of AL , which is $g_A + g_L$, is plotted in the dashed line. Everywhere the dashed line is above the solid line, the $K_t/A_t L_t$ ratio must be falling, as $g_K < g_A + g_L$. Everywhere the dashed line is below the solid line, the $K_t/A_t L_t$ ratio must be rising, as $g_K > g_A + g_L$.

Use a phase diagram

Figure 5.1 The Dynamics of Productivity Growth



NOTE: The two curves plot the growth rate of the numerator and denominator of the ratio $L_t^\lambda / A_t^{1-\phi}$. The growth rate of the numerator is λg_L , plotted as the horizontal line. The growth rate of the denominator is $(1-\phi)g_A$, which from equation (5.8) is increasing in the ratio. The economy reaches a steady state where the two lines intersect, and at that point the growth rate of productivity is $g_A^{ss} = \frac{\lambda}{1-\phi} g_L$.

Characterizing the steady state

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$$\left(\frac{L^\lambda}{A^{1-\phi}} \right)^{\text{ss}} = \frac{g_A^{\text{ss}}}{\theta s_R^\lambda},$$

which means, along the BGP,

$$A_t^{\text{BGP}} = \left(\frac{\theta s_R^\lambda}{g_A^{\text{ss}}} L_t^\lambda \right)^{\frac{1}{1-\phi}}$$

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Let $\tilde{k}_t \equiv K_t / (A_t(1 - s_R)L_t) \rightarrow$ notice s_R !

Output per capita along the BGP is then

$$\begin{aligned} y_t^{\text{BGP}} &= \left(\tilde{k}^{\text{ss}} \right)^\alpha (1 - s_R) A_t^{\text{BGP}} \\ &= \left(\tilde{k}^{\text{ss}} \right)^\alpha (1 - s_R) \left(\frac{\theta s_R^\lambda}{g_A^{\text{ss}}} \right)^{\frac{1}{1-\phi}} L_t^{\frac{\lambda}{1-\phi}} \end{aligned}$$

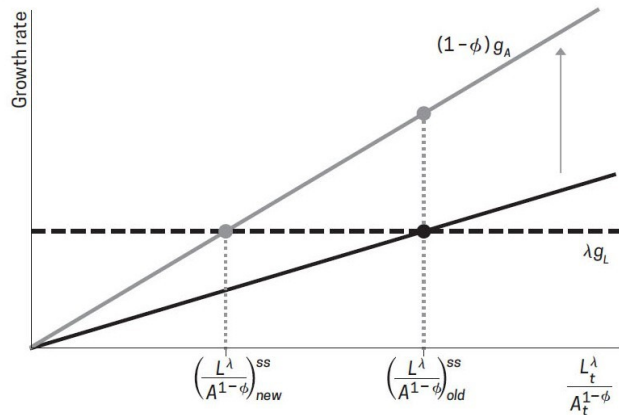
where, as in the Solow model,

$$\tilde{k}^{\text{ss}} = \left(\frac{s_I}{g_A^{\text{ss}} + g_L + \delta} \right)^{\frac{1}{1-\alpha}},$$

but now g_A^{ss} is **endogenous**!

An increase in s_R : What happens to $L_t^\lambda / A_t^{1-\phi}$?

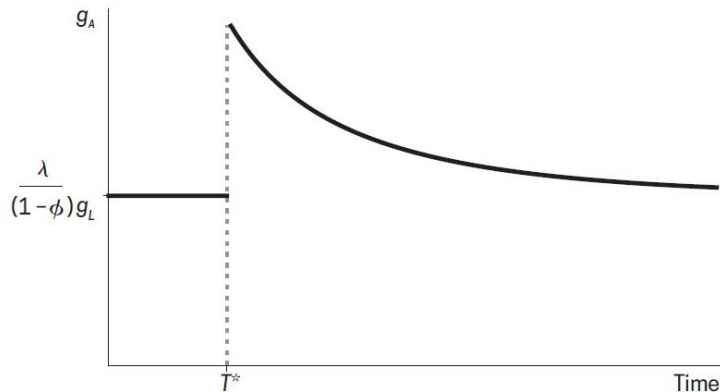
Figure 5.2 The Dynamics of an Increase in s_R



NOTE: An increase in s_R to s'_R rotates the g_A line upwards. In response, the growth rate g_A jumps up (the gray dot) and the ratio $L_t^\lambda / A_t^{1-\phi}$ declines to the new steady state. At that point the steady state growth rate of productivity is again $g_A^{ss} = \lambda g_L / (1 - \phi)$.

An increase in s_R : What happens to g_A ?

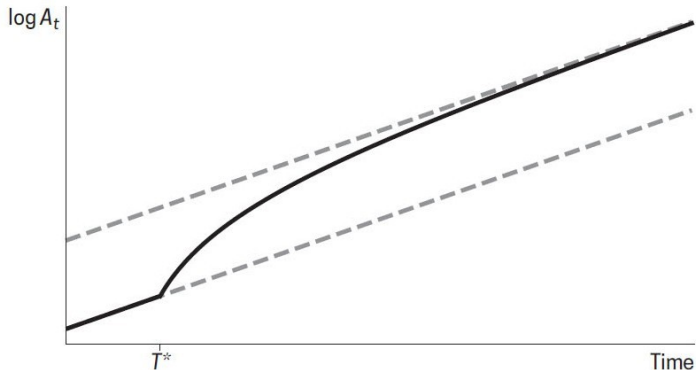
Figure 5.3 The Growth Rate of Productivity over Time



NOTE: At time T^* the share of researchers, s_R , increases. The growth rate spikes above the steady state value due to the increased production of ideas. But eventually the growth returns to the long-run value. Note that the steady state growth rate $g_A^{ss} = \lambda g_L / (1 - \phi)$.

An increase in s_R : What happens to $\ln A_t$?

Figure 5.4 The Level of Productivity over Time



NOTE: At time T^* the share of researchers, s_R , increases. Because the growth rate g_A is temporarily higher, the level of productivity, $\log A_t$, increases rapidly and then remains along a higher path even as g_A falls back to the steady state growth rate. The dashed lines indicate the hypothetical paths of the lower and higher level of productivity the economy is moving between.

An increase in s_R : What happens to $\ln y_t$?

Recall...

$$\begin{aligned}\ln y_t &= \ln(1 - s_R) + \ln A_t + \alpha \ln \tilde{k}_t \\ &= (1 - \alpha) [\ln(1 - s_R) + \ln A_t] + \alpha \ln k_t\end{aligned}$$

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... and therefore

$$\begin{aligned}g_y &= g_A + \alpha(g_K - g_A - g_L) \\ &= (1 - \alpha)g_A + \alpha(g_K - g_L)\end{aligned}$$

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with, in the long run, ...

$$\begin{aligned}g_y^{\text{ss}} &= g_A^{\text{ss}} = \frac{\lambda}{1 - \phi} g_L \\ \tilde{k}^{\text{ss}} &= \left(\frac{s_I}{g_A^{\text{ss}} + g_L + \delta} \right)^{\frac{1}{1 - \alpha}}\end{aligned}$$

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$$\begin{aligned}g_y^{ss} &= g_A^{ss} = \frac{\lambda}{1 - \phi} g_L \\ \tilde{k}^{ss} &= \left(\frac{s_I}{g_A^{ss} + g_L + \delta} \right)^{\frac{1}{1 - \alpha}}\end{aligned}$$

So, at the time of the shock (T^*)...

- g_A immediately *jumps up*
- $\ln A_t$ does not immediately change
- $\ln y_t$ *drops* even though $\tilde{k}_t$ *jumps up*
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An increase in s_R : What happens to $\ln y_t$?

Recall...

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But in between?

- generically, faster when farther
- but some **drag** from $\tilde{k}_t$ adjusting

What values for λ and ϕ can rationalize the data?

Consider the original Romer (1990) case: $\lambda = \phi = 1$

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 - growth rate of productivity depends on the **level** of population
 - growth would **explode** if population growth was positive

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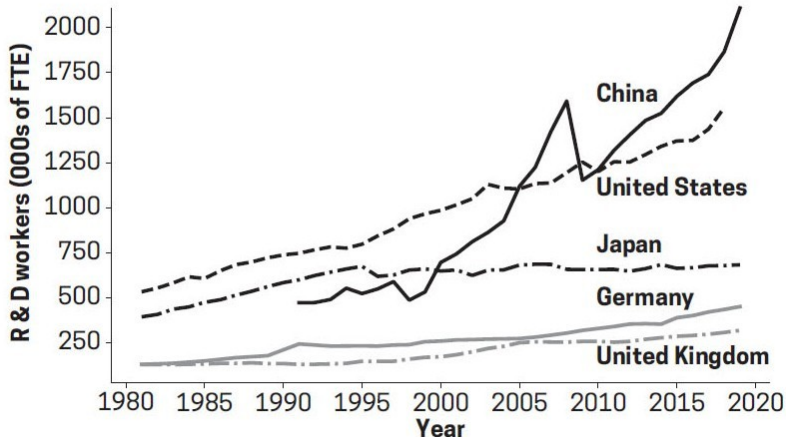
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 - would suggest 10 people can produce as many ideas as 10mil if same s_R
 - in the data, even s_R has been growing!
- both imply a **growth effect** of changing s_R (just saw **level effect** when $\phi < 1$)

R&D employment *level* has been rising. . .

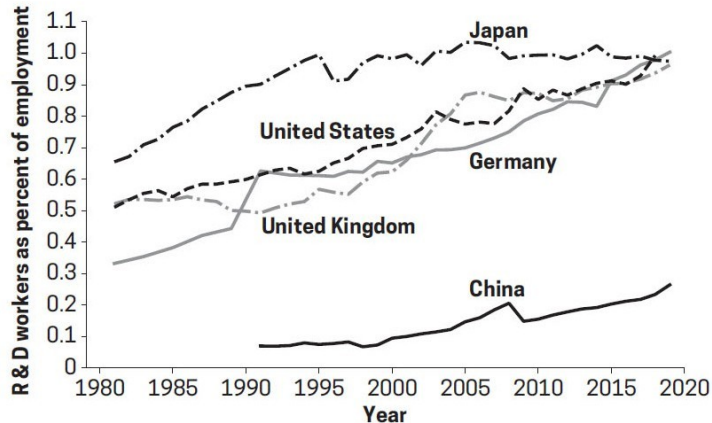
Figure 4.2 Number of R&D Workers (FTE), by Country



SOURCE: OECD Research and Development Statistics (2019).

... as has its *share*

Figure 4.3 R&D Workers as a Percent of Employment,
by Country



SOURCE: Authors' calculations from OECD Research and Development Statistics (2019).

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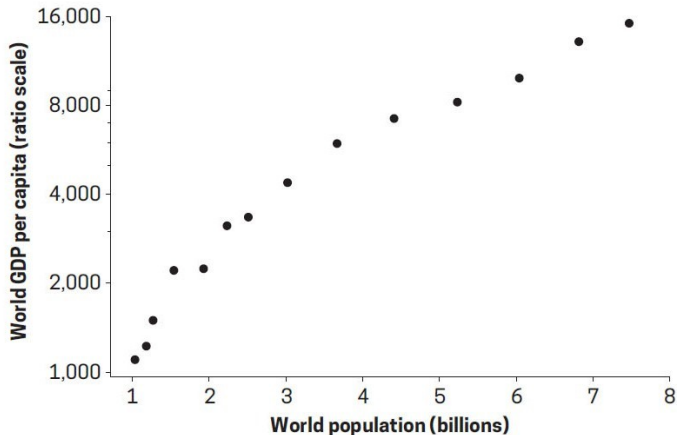
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- **why?** more people don't dilute “ideas per person” because ideas are **nonrival**
(vs. capital in NGM)
- **but is that implication borne out in the data?**
 1. should only think about applying this model to the most advanced economies
 2. even among advanced economies, technology diffusion muddies the waters
 3. the model actually relates the growth in the *effective number of researchers* ($s_R L$), rather than population, to economic growth → more growth in former than latter

World population and world GDP per capita have moved hand-in-hand

Figure 4.6 World Population and GDP per Capita, 1820–2019



SOURCE: Authors' calculations from Bolt and van Zanden (2020).

- ideas are special kind of economic good

Ideas → Nonrivalry → Increasing returns → Imperfect competition

Summary so far

- ideas are special kind of economic good

Ideas $\rightarrow$ Nonrivalry $\rightarrow$ Increasing returns $\rightarrow$ Imperfect competition

- build (semi-)endogenous growth model w/ costly, deliberate idea accumulation
 - started w/ aggregate dynamics in which A grows endogenously...
 - **next**: what economic choices determine s_R in Romer (1990)?

An overview of production

There are two layers of production:

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1. set of **final good** producers in a **perfectly competitive** market
 - think about Walmart, Target, ...
 - combine labor and intermediate goods to produce the final good
 - take prices as given (normalize output price to 1)

An overview of production

There are two layers of production:

1. set of **final good** producers in a **perfectly competitive** market
 - think about Walmart, Target, ...
 - combine labor and intermediate goods to produce the final good
 - take prices as given (normalize output price to 1)
2. set of **intermediate good** producers in a **monopolistically competitive** market
 - sell the inventory, cash registers, buildings, etc. to Walmart
 - just use capital to produce each intermediate good (simplifying, not necessary)
 - each intermediate good is “small” but unique → have some **market power**
 - will **choose** profit-maximizing price

The final goods sector

Assume a large number of identical firms using the same Cobb-Douglas technology...

$$Y_t = L_{Yt}^{1-\alpha} \int_0^{A_t} x_{jt}^\alpha dj \quad (L_{Yt} = (1 - s_R)L_t)$$

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... hire labor and intermediate goods to **maximize profits**:

$$\max_{L_{Yt}, \{x_{jt}\}} L_{Yt}^{1-\alpha} \int_0^{A_t} x_{jt}^\alpha dj - wL_{Yt} - \int_0^{A_t} p_{jt}x_{jt} dj,$$

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which has first-order conditions

$$w_t = (1 - \alpha) \frac{Y_t}{L_{Yt}}, \quad p_{jt} = \alpha L_{Yt}^{1-\alpha} x_{jt}^{\alpha-1} \text{ for all } j \in [0, \mathbf{A}_t]$$

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There are A_t monopolistically competitive firms, each of which. . .

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- ... in order to **maximize (operating) profits**

$$\max_{x_{jt}} \pi_{jt} = p_{jt}(x_{jt}) x_{jt} - r x_{jt}$$

Profit maximization for intermediate goods producer

The first-order condition is

$$\boldsymbol{r} = p_{jt} + x_{jt} \frac{\partial p_{jt}}{\partial x_{jt}} = p_{jt} \left(1 + \frac{x_{jt}}{p_{jt}} \frac{\partial p_{jt}}{\partial x_{jt}} \right) = p_{jt} (1 + (\alpha - 1)) = \boldsymbol{\alpha} p_{jt}$$

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...so $\alpha \rightarrow 1$ looks like perfect competition!

Adding up production

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Intermediate producers set the same price $\rightarrow$ choose the same quantity $x_{jt} = x_t$

$$\int_0^{A_t} x_{jt} dj = K_t \quad \rightarrow \quad x_t = \frac{K_t}{A_t}$$

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Substituting this back into the final goods production function...

$$Y_t = L_{Yt}^{1-\alpha} A_t x_t^\alpha = L_{Yt}^{1-\alpha} A_t \left(\frac{K_t}{A_t} \right)^\alpha = K_t^\alpha (A_t L_{Yt})^{1-\alpha}$$

... which is where we started before introducing the microfoundation!

Profits and their growth

How much do final goods producers **spend** on intermediates?

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It follows that along a BGP...

$$g_\pi = g_Y - g_A = (g_A + g_L) - g_A = g_L$$

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- A (potential) producer will try to create a new idea if $F_t \leq V_t$
- **Free entry** of potential producers → competition ensures $F_t = V_t$ in equilibrium

Putting it all together to pin down s_R on BGP

Start from the free entry condition $F_t = V_t$:

$$\frac{\pi_t}{r - g_\pi} = w_t \frac{s_R L_t}{\dot{A}_t}$$

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$$\frac{\alpha(1 - \alpha) \frac{Y_t}{\dot{A}_t}}{r - g_L} = \left[(1 - \alpha) \frac{Y_t}{(1 - s_R) L_t} \right] \frac{s_R L_t}{g_A \dot{A}_t}$$

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Rearrange and cancel terms:

$$\frac{s_R}{1 - s_R} = \frac{\alpha(1 - \alpha)}{1 - \alpha} \frac{g_A}{r - g_L} \rightarrow s_R = \frac{1}{1 + \frac{r - g_L}{\alpha g_A}}$$

(r , g_A are endogenous but independent of s_R on BGP)

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